

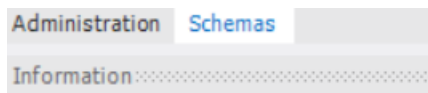
MYSQL ASSIGNMENT 1 - DDL COMMANDS, CONSTRAINTS

1. Database and Table Creation (CREATE):

- Create a Database for this project?

Database Name: Employee Database

Screenshots:



Schema: **employee_database**

- Create all three tables in MySQL with appropriate **data types** and **relationships**?

Table: Departments

Screenshot:

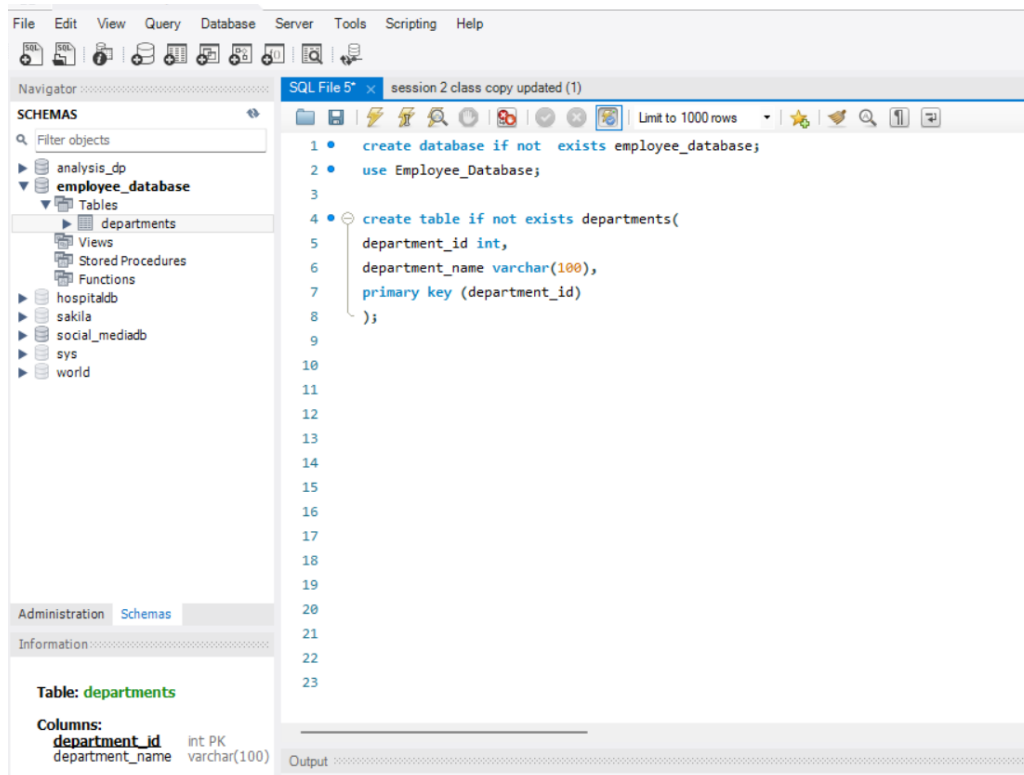


Table: Location

Screenshot:

The screenshot displays the SQL Server Enterprise Manager interface on the left and the SQL Server Enterprise Edition query editor on the right.

SQL Server Enterprise Manager (Left Panel):

- Navigator:** Shows the database hierarchy. The **employee_database** is expanded, showing **Tables** (departments, location), **Views**, **Stored Procedures**, and **Functions**. The **location** table is selected.
- Administration:** Shows the **Schemas** tab.
- Information:** Displays details for the **Table: location**.
 - Columns:**
 - location_id**: int PK
 - location_name**: varchar(100)

SQL Server Enterprise Edition (Right Panel):

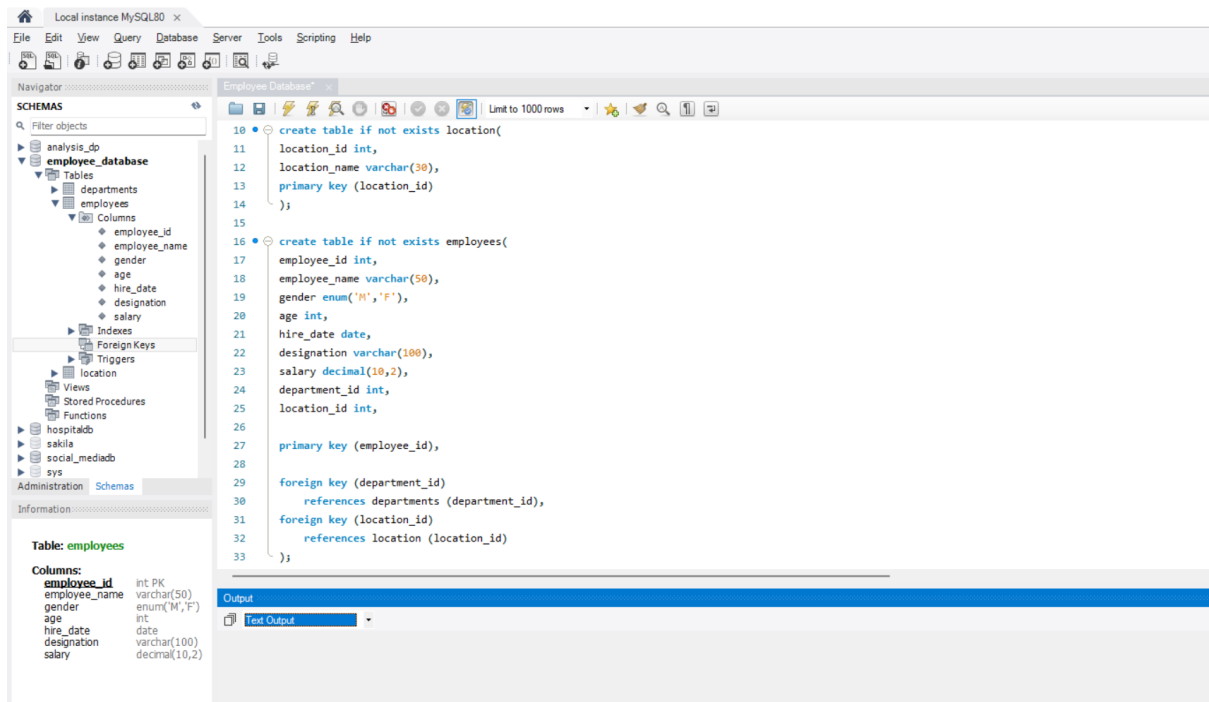
The query editor shows the following SQL code:

```
1 • create database if not exists employee_database;
2 • use Employee_Database;
3
4 • create table if not exists departments(
5     department_id int,
6     department_name varchar(100),
7     primary key (department_id)
8 );
9
10 • create table if not exists location(
11     location_id int,
12     location_name varchar(100),
13     primary key (location_id)
14 );
15
16
17
18
19
20
21
22
23
24
```

The **Output** pane at the bottom is empty.

Table: Employees

Screenshot:



2. Table Alteration (ALTER)

- Add a new column named "email" to the Employees table to store employee email addresses.

Screenshot:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Schemas' pane displays the 'employee_database' schema with its tables: 'departments', 'employees', and 'location'. The 'employees' table is selected, and its columns are listed: 'employee_id' (int PK), 'employee_name' (varchar(50)), 'gender' (enum('M','F')), 'age' (int), 'hire_date' (date), 'designation' (varchar(100)), 'salary' (decimal(10,2)), and 'email' (varchar(100)).

The main pane shows a SQL script for the 'employees' table:

```

26
27
28 primary key (employee_id),
29
30 foreign key (department_id)
31 references departments (department_id),
32 foreign key (location_id)
33 references location (location_id)
34 );
35
36 alter table employees
37 add column email varchar(100);
38
39
40
41
42
43
44
45
46
47
48
49

```

The bottom pane shows the 'Output' window with the following messages:

#	Time	Action	Message
8	05:37:04	create table if not exists employees(employee_id int...	Error Code: 1064. You have an error in yc
9	05:37:24	create table if not exists employees(employee_id int...	0 row(s) affected, 1 warning(s): 1050 Tab
10	05:37:49	foreign key (department_id) references departments ...	Error Code: 1064. You have an error in yc
11	05:40:19	foreign key (department_id) references departments ...	Error Code: 1064. You have an error in yc
12	05:42:56	create table if not exists employees(employee id int...	0 row(s) affected, 1 warning(s): 1050 Tab

- Modify the data type of the "designation" column in the Employees table to support a wider range of values.

Screenshot:

Navigator

Employee Database

Limit to 1000 rows

20 age int,
 21 hire_date date,
 22 designation varchar(100),
 23 salary decimal(10,2),
 24 department_id int,
 25 location_id int,
 26
 27 primary key (employee_id),
 28
 29 foreign key (department_id)
 30 references departments (department_id),
 31 foreign key (location_id)
 32 references location (location_id)
 33);
 34
 35 • alter table employees
 36 add column email varchar(100);
 37
 38 • alter table employees
 39 modify column designation varchar(255);
 40
 41
 42
 43

SCHEMAS

Filter objects

analysis_dp

employee_database

Tables

departments

employees

Columns

employee_id

employee_name

gender

age

hire_date

designation

salary

email

Indexes

Foreign Keys

Triggers

location

Views

Stored Procedures

Functions

hospitaldb

sakila

social_mediadb

Administration

Schemas

Information

Table: employees

Columns:

employee_id int PK

employee_name varchar(50)

gender enum('M','F')

age int

hire_date date

designation varchar(255)

salary decimal(10,2)

email varchar(100)

Output

Action Output

#	Time	Action	Message
13	05:54:00	alter table employees add column email varchar(100)	0 row(s) affected Records: 0 Duplicate
14	05:58:16	alter table employees modify column designation var...	Error Code: 1146. Table 'employee_dat

- Drop the “age” column from the Employees table.

Screenshot:

Table: employees

Columns:

Column Name	Data Type
employee_id	int PK
employee_name	varchar(50)
gender	enum('M','F')
hire_date	date
designation	varchar(255)
salary	decimal(10,2)
email	varchar(100)

```

20 age int,
21 hire_date date,
22 designation varchar(100),
23 salary decimal(10,2),
24 department_id int,
25 location_id int,
26
27 primary key (employee_id),
28
29 foreign key (department_id)
30 references departments (department_id),
31 foreign key (location_id)
32 references location (location_id)
33 );
34
35 • alter table employees
36 add column email varchar(100);
37
38 • alter table employees
39 modify column designation varchar(255);
40
41 • alter table employees
42 drop column age;
43

```

Output

#	Time	Action
14	05:58:16	alter table employees modify column designation varchar(100)

- Rename the “hire_date” column to “date_of_joining”.

Screenshot:

Table: employees

Columns:

Column Name	Data Type	Constraints
employee_id	int	PK
employee_name	varchar(50)	
gender	enum('M','F')	
date_of_joining	date	
designation	varchar(255)	
salary	decimal(10,2)	
email	varchar(100)	

Foreign Keys:

- Primary Key: employee_id
- Foreign Key: department_id (references departments (department_id))
- Foreign Key: location_id (references location (location_id))

SQL Commands:

```

26
27 primary key (employee_id),
28
29 foreign key (department_id)
30     references departments (department_id),
31 foreign key (location_id)
32     references location (location_id)
33 );
34
35 • alter table employees
36   add column email varchar(100);
37
38 • alter table employees
39   modify column designation varchar(255);
40
41 • alter table employees
42   drop column age;
43
44 • alter table employees
45   rename column hire_date to date_of_joining;
46
47
48
49

```

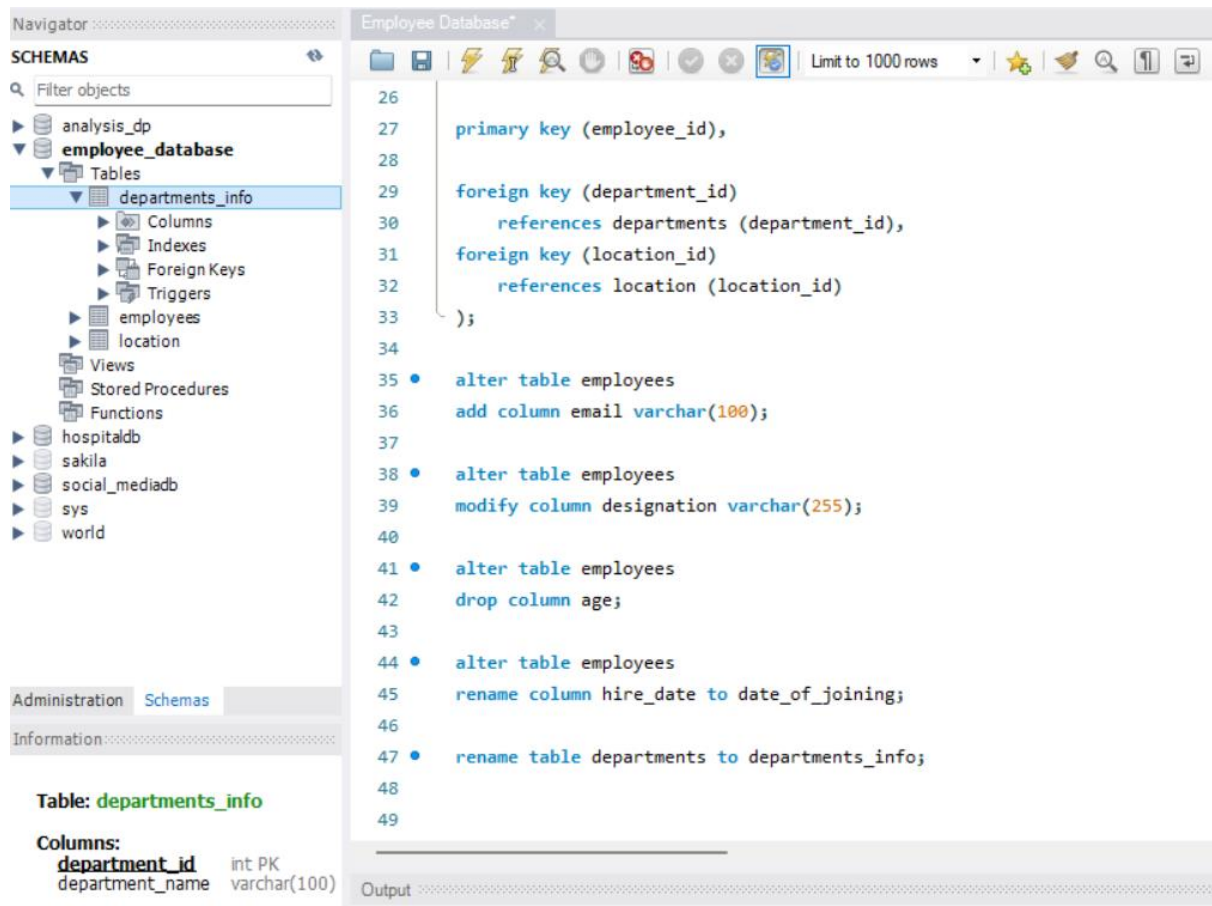
Output:

#	Time	Action	Message
15	05:58:30	alter table employees modify column designation var...	0 row(s) affected Records: 0 Duplicates: 0
16	06:01:29	alter table employees add column email varchar(255);	Error Code: 1060. Duplicate column name 'email'.

3. Table Renaming (RENAME):

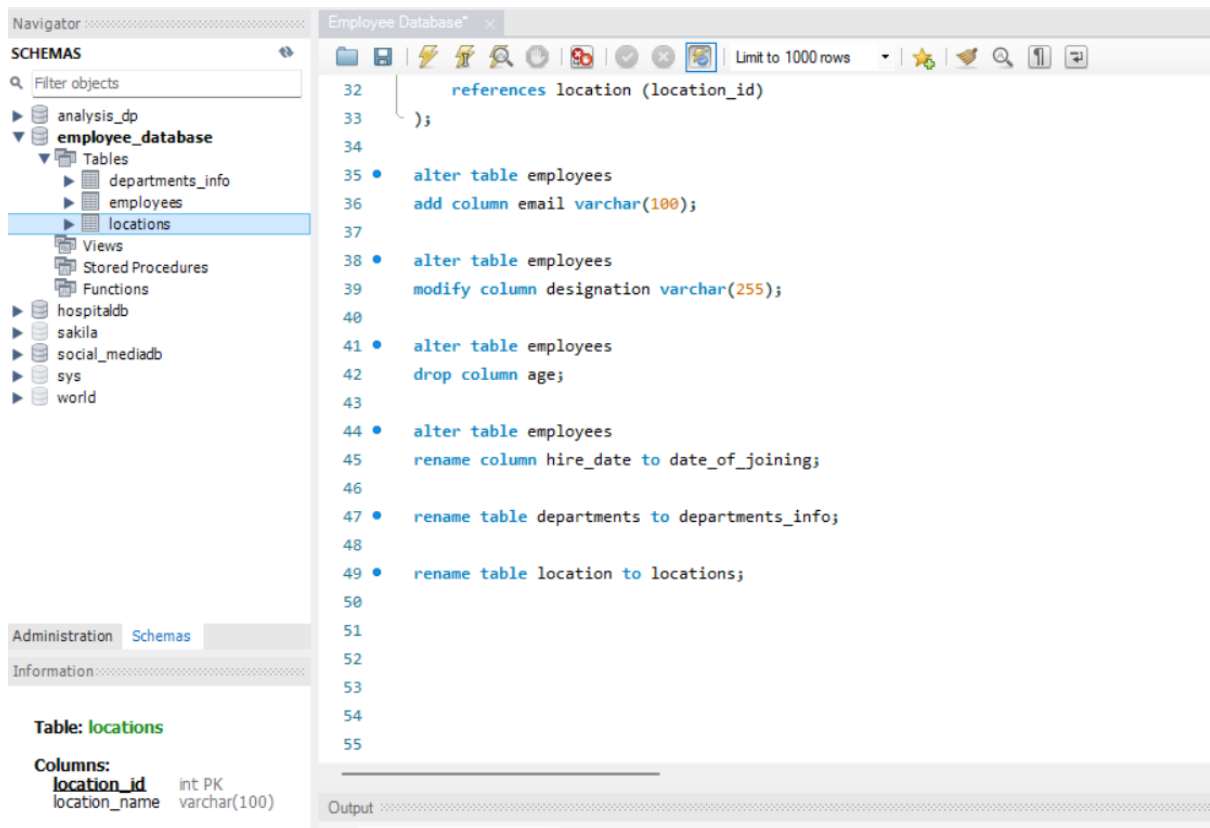
- Rename the "Departments" table to "Departments_Info".

Screenshot:



- Rename the "Location" table to "Locations".

Screenshot:



4. Table Truncation (TRUNCATE):

- Truncate the Employees table.

Screenshot:

The screenshot displays a database management interface with a left-hand sidebar and a main workspace.

Left Sidebar:

- Navigator:** Shows a tree view of schemas. The **employee_database** schema is expanded, showing tables: **departments_info**, **employees**, and **locations**. Below these are **Columns**, **Indexes**, **Foreign Keys**, and **Triggers**. Other schemas listed include **analysis_dp**, **hospitaladb**, **sakila**, **social_medadb**, **sys**, and **world**.
- Administration / Schemas:** A tabbed interface with **Schemas** selected.
- Information:** A section titled **Table: employees** showing column details:
 - Columns:** **employee_id** (int PK), **employee_name** (varchar(50)), **gender** (enum('M','F')), **date_of_joining** (date), **designation** (varchar(255)), **salary** (decimal(10,2)), **email** (varchar(100)).
- Object Info / Session:** A tabbed interface with **Session** selected.

Main Workspace:

- SQL Editor:** Contains a series of SQL queries:


```

32 references location (location_id)
33 );
34
35 • alter table employees
36 add column email varchar(100);
37
38 • alter table employees
39 modify column designation varchar(255);
40
41 • alter table employees
42 drop column age;
43
44 • alter table employees
45 rename column hire_date to date_of_joining;
46
47 • rename table departments to departments_info;
48
49 • rename table location to locations;
50
51 • truncate table employees;
52
53
54
55
      
```
- Output:** A section titled **Action Output** showing the results of the executed queries:

#	Time	Action	Message
✓ 18	06:04:31	alter table employees modify column designation var...	0 row(s) affected Records: 0 Duplicates: 0 Wamin...
✓ 19	06:09:46	alter table employees drop column age	0 row(s) affected Records: 0 Duplicates: 0 Wamin...
✓ 20	06:14:12	alter table employees rename column hire_date to d...	0 row(s) affected Records: 0 Duplicates: 0 Wamin...
✓ 21	06:19:39	rename table departments to departments_info	0 row(s) affected
✓ 22	06:21:30	rename table location to locations	0 row(s) affected
✓ 23	06:25:56	truncate table employees	0 row(s) affected

5. Database & Table Dropping (DROP):

- Drop the Employees table and then the “employee” database.

Screenshots:

Navigator

SCHEMAS

Filter objects

- analysis_dp
- employee_database
 - Tables
 - departments_info
 - locations
 - Views
 - Stored Procedures
 - Functions
- hospitaldb
- sakila
- social_medadb
- sys
- world

Administration Schemas

Information

No object selected

Employee Database*

Limit to 1000 rows

```

32     references location (location_id)
33 );
34
35 • alter table employees
36   add column email varchar(100);
37
38 • alter table employees
39   modify column designation varchar(255);
40
41 • alter table employees
42   drop column age;
43
44 • alter table employees
45   rename column hire_date to date_of_joining;
46
47 • rename table departments to departments_info;
48
49 • rename table location to locations;
50
51 • truncate table employees;
52
53 • drop table employees;
54
55

```

Output

Action Output

#	Time	Action	Message	Duration / Fets
19	06:09:46	alter table employees drop column age	0 row(s) affected Records: 0 Duplicates: 0 Wamin...	0.094 sec
20	06:14:12	alter table employees rename column hire_date to d...	0 row(s) affected Records: 0 Duplicates: 0 Wamin...	0.078 sec
21	06:18:38	rename table departments to departments_info	0 row(s) affected	0.003 sec

Navigator

SCHEMAS

Filter objects

- analysis_dp
- hospitaldb
- sakila
- social_medadb
- sys
- world

Administration Schemas

Information

No object selected

Employee Database*

Limit to 1000 rows

```

34
35 • alter table employees
36   add column email varchar(100);
37
38 • alter table employees
39   modify column designation varchar(255);
40
41 • alter table employees
42   drop column age;
43
44 • alter table employees
45   rename column hire_date to date_of_joining;
46
47 • rename table departments to departments_info;
48
49 • rename table location to locations;
50
51 • truncate table employees;
52
53 • drop table employees;
54
55 • drop database employee_database;
56
57

```